ONCOLOGY · NEET-SS / FET / INI-SS

Cartilage tumours

1. Overview

- **Chondrogenic tumours** form hyaline cartilage matrix. Radiographic hallmark of cartilage matrix: **"rings and arcs" / "popcorn" / punctate (stippled) calcification**.
- **WHO 2020 classification (bone):**
  - **Benign:** osteochondroma, enchondroma, periosteal chondroma, chondroblastoma, chondromyxoid fibroma, subungual exostosis, bizarre parosteal osteochondromatous proliferation (BPOP), synovial chondromatosis, osteochondromyxoma.
  - **Intermediate (locally aggressive): atypical cartilaginous tumour (ACT)** of the appendicular skeleton (formerly grade 1 chondrosarcoma).
  - **Malignant:** chondrosarcoma grades 1-3 (grade 1 term kept for **axial skeleton**: pelvis, scapula, skull base), periosteal chondrosarcoma, **clear cell**, **mesenchymal** and **dedifferentiated** chondrosarcoma.
- **Location in bone (classic exam table):**

Location	Cartilage lesion
Epiphysis, skeletally immature	**Chondroblastoma**
Epiphysis, adult	**Clear cell chondrosarcoma** (and GCT, which is not cartilaginous)
Metaphysis, eccentric	**Chondromyxoid fibroma**, osteochondroma
Medullary metaphysis/diaphysis	Enchondroma, central chondrosarcoma

Location	Cartilage lesion
Surface	Osteochondroma, periosteal chondroma, periosteal chondrosarcoma
Short tubular bones of the hand	Enchondroma (commonest primary bone tumour of the hand)

- **Molecular high-yield: IDH1/IDH2 mutations** in enchondroma, Ollier, Maffucci and about 50% of central (conventional and dedifferentiated) chondrosarcomas; **EXT1/EXT2** in osteochondroma and HME; **H3F3B K36M (H3.3)** in chondroblastoma; **GRM1** rearrangement in chondromyxoid fibroma; **HEY1-NCOA2** fusion in mesenchymal chondrosarcoma.

2. Osteochondroma (osteocartilaginous exostosis)

Pathology and epidemiology

- **Commonest benign bone tumour** (about one third of benign bone tumours; many unrecognised). Age 10-30, slight male predominance.
- A **developmental lesion**: physeal cartilage herniates through the periosteal bone collar (perichondral ring of LaCroix) and grows at right angles to the bone.
- **Defining feature: cortical and medullary continuity with the host bone**, capped by hyaline cartilage. It grows by enchondral ossification under the cap and **stops growing at skeletal maturity**.
- **Sites:** metaphysis of long bones, especially **distal femur, proximal tibia (around the knee) and proximal humerus**; points **away from the adjacent joint** (grows along tendon pull). Sessile or pedunculated.
- Radiation can induce osteochondromas (e.g. after childhood irradiation).

Clinical features and complications

- Painless hard mass; pain from **overlying bursa (bursitis)**, tendon snapping, **fracture of the stalk**.
- **Neurovascular compression: common peroneal nerve** at the proximal fibula; **popliteal artery pseudoaneurysm** or thrombosis from distal femoral or proximal tibial lesions.
- Growth disturbance and deformity (especially in multiple exostoses), spinal cord compression (rare spinal lesions).
- **Malignant transformation to secondary peripheral chondrosarcoma: about 1% for solitary lesions**; **about 1-5% (lifetime) in HME** (older figures up to 10-20% are overestimates; debated). Pelvis, scapula and proximal femur lesions are the highest risk.
- **Warning signs of malignancy: growth or new pain after skeletal maturity**, **cartilage cap thicker than 2 cm in an adult** (some use 1.5 cm) on MRI, irregular cap with lucent areas or soft tissue mass with scattered calcification, cortical destruction of the stalk.
  - In growing children, caps up to 3 cm can be normal.

Treatment

- **Observation** for asymptomatic lesions.
- **Excision** for pain, neurovascular compression, cosmesis, deformity or suspicion of malignancy: **remove the whole cartilage cap with its perichondrium** (and stalk base) to prevent recurrence. Ideally after skeletal maturity to reduce recurrence and avoid physeal damage.
- **Peroneal nerve** must be identified and protected when excising proximal fibular lesions.
- Suspected malignant change: biopsy-guided **wide en bloc resection** (do not shell out).

Hereditary multiple exostoses (HME, multiple osteochondromas)

- **Autosomal dominant**, high penetrance (about 96%); **EXT1 (chromosome 8q24)** gives a more severe phenotype than **EXT2 (11p)**. EXT genes encode glycosyltransferases for **heparan sulphate** synthesis.
- **Short stature**, limb length discrepancy, **genu valgum** (proximal tibia), **ankle valgus** (distal tibia and fibular shortening), **coxa valga** and hip subluxation.
- **Forearm deformity (in 40-60%)**: the **ulna is short** (lesions at the distal ulna), radius bowed with ulnar inclination of the distal radius, ulnar translation of the carpus, and **radial head dislocation**.
- **Masada classification of the forearm:**

Type	Description	Management idea
I	Osteochondroma of distal ulna; ulna short; radius bowed; radial head **not** dislocated	Excision + ulnar lengthening (gradual distraction) ± radial corrective osteotomy
IIa	Radial head dislocated, with an osteochondroma of the proximal radius	Excision; radial head excision at maturity if painful
IIb	Radial head dislocated secondary to a short ulna (distal ulnar lesion), no proximal radial lesion	Ulnar lengthening ± open reduction of radial head (difficult)
III	Osteochondroma of distal radius with relative shortening of the radius	Excision

- **Treatment principles in HME:** excise symptomatic lesions, **guided growth (hemiepiphysiodesis)** for genu valgum and ankle valgus, lengthening for forearm and limb discrepancy, lifelong awareness of malignant change (new pain or growth in adulthood).
- **Related conditions:**
  - **Trevor disease (dysplasia epiphysealis hemimelica):** intra-articular **epiphyseal osteochondroma**, usually on the **medial**

side of one lower limb (talus, distal tibia, distal femur); boys about 3:1. Excise symptomatic lesions; preserve articular surface.

- **Metachondromatosis:** AD, **PTPN11**; both exostoses (which **point towards the joint**, mainly hands and feet) and enchondromas; lesions may regress.
- **Subungual exostosis (Dupuytren):** distal phalanx of the great toe, trauma-related, **no medullary continuity**; excise with cap.
- **BPOP (Nora lesion):** surface lesion of hands and feet, no corticomedullary continuity, high recurrence.

Exam pearl: Osteochondroma = corticomedullary continuity + cartilage cap, points away from the joint, stops growing at maturity. Cap over 2 cm in an adult or growth/pain after maturity = suspect chondrosarcoma. HME = AD, EXT1 (8q24, worse) / EXT2 (11p), heparan sulphate, short ulna with radial head dislocation (Masada).

3. Enchondroma and enchondromatoses

Solitary enchondroma

- Benign intramedullary hyaline cartilage, probably from physeal rests. Age 20-50. Often an incidental finding.
- **Commonest primary bone tumour of the hand** (about half occur in the hand): **proximal phalanx** most common, then metacarpals and middle phalanges. Also proximal humerus, distal femur and proximal tibia (metaphysis to diaphysis).
- **Hand:** presents with **pathological fracture** after minor trauma; expansile lytic lesion with thinned cortex and stippled calcification.
  - **Management:** allow the fracture to heal (about 4-6 weeks), then **curettage ± bone graft** or bone substitute (some treat in one stage). Recurrence is uncommon. Malignant change in the hand is very rare.

- **Long bones:** central lesion with **rings-and-arcs calcification**, little or no endosteal scalloping, no cortical breach, no soft tissue mass, usually painless.
  - Painless lesions without aggressive features: **observation with serial imaging** (radiographs or MRI).

Enchondroma versus atypical cartilaginous tumour / low-grade chondrosarcoma in long bones

Feature	Favours enchondroma	Favours ACT / chondrosarcoma
Pain	Absent (pain usually from another cause, e.g. knee OA)	**Pain attributable to the lesion**, especially at night
Endosteal scalloping	Absent or shallow	**Deep (more than two thirds of cortical thickness)** and long (over two thirds of lesion length)
Cortex	Normal	**Cortical thickening, expansion, breach**
Periosteal reaction / soft tissue mass	Absent	Present
Size	Usually under 5 cm	Larger (often over 5 cm)
Bone scan uptake	Mild	Greater than the anterior iliac crest
Site	Hand, long bones	**Axial skeleton (pelvis) almost always malignant**
Histology	Hypocellular lobules **encasing** host trabeculae	**Permeation and entrapment of host trabeculae**, myxoid change, increased cellularity

- Histology alone is unreliable for distinguishing enchondroma from low-grade chondrosarcoma (high interobserver variability); **clinical, radiological and pathological correlation** is required.
- **Bone infarct** (differential): **serpiginous peripheral sclerotic rim** ("smoke rising up a chimney"), no scalloping, no central rings and arcs.

Ollier disease and Maffucci syndrome

Condition	Features	Malignancy risk
Ollier disease (enchondromatosis)	**Multiple enchondromas**, often **unilateral predominance**, sporadic (not inherited), **somatic mosaic IDH1/IDH2** mutations; limb shortening, angular deformity, short ulna with forearm bowing, hand deformity	**Secondary central chondrosarcoma in about 25-40%** (figures vary; debated)
Maffucci syndrome	Multiple enchondromas + **soft tissue (spindle cell) haemangiomas** (blue nodules with **phleboliths** on radiographs); IDH1/IDH2	**Higher (up to about 50%)**, plus **non-skeletal malignancies** (gliomas, ovarian juvenile granulosa cell tumour, pancreatic and hepatic tumours)

- Lifelong surveillance; any new pain or growth needs MRI and evaluation.
- Limb deformity: corrective osteotomies, gradual lengthening (Ilizarov / hexapod); lesions within distraction regenerate ossify.

Periosteal (juxtacortical) chondroma

- Surface cartilage lesion under the periosteum, young adults, **proximal humerus** commonest, hand.
- **Saucerisation of the cortex with a sclerotic buttress**, no medullary involvement; usually **under 3 cm**.
- Differential: **periosteal chondrosarcoma** (larger, usually over 3-5 cm, older patients, invasion of cortex/medulla).
- Treatment: **marginal en bloc excision** with the underlying cortex; low recurrence.

4. Chondroblastoma

- About 1% of bone tumours. **Age 10-25 (skeletally immature)**; male predominance about 2:1.
- **Epiphysis or apophysis** (equivalent): **proximal humerus (Codman tumour)**, **distal femur**, **proximal tibia**, **greater trochanter**; also **talus and calcaneus** in older patients.
- **Symptoms:** joint pain, effusion, stiffness (mimics intra-articular pathology).
- **Radiograph:** small, round, well-defined **lytic epiphyseal lesion with a thin sclerotic rim**, central calcification in about a third to half; may cross the physis into the metaphysis.
- **MRI: extensive surrounding marrow and soft tissue oedema** and periosteal reaction (out of proportion to the lesion); **secondary ABC** change in about 15-30% (fluid-fluid levels).
- **Histology:** polygonal **chondroblasts with grooved "coffee-bean" nuclei**, scattered osteoclast-like giant cells, pink chondroid matrix, **"chicken-wire" pericellular calcification**. Immunohistochemistry: S100, SOX9, DOG1 and **H3 K36M mutant-specific antibody** (H3F3B).
- **Treatment: extended curettage** (high-speed burr ± phenol, cryotherapy or argon adjuvant) and grafting or cementing, approached to protect the **physis and articular cartilage** (e.g. through the metaphysis or via a cortical window). **Local recurrence**

about 10-20% (higher in pelvis, hip, talus, and with secondary ABC). RFA for small lesions in difficult sites.

- **Benign lung metastases** in about 1-2% (usually after local recurrence); treated by excision, good prognosis.
- **Differentials (epiphyseal lucent lesions):** GCT (skeletally mature, no sclerotic rim, extends to subchondral bone), clear cell chondrosarcoma (adult), infection (Brodie abscess can be epiphyseal in children), LCH, intraosseous ganglion, subchondral cyst/geode.

5. Chondromyxoid fibroma (CMF)

- **Rarest benign cartilage tumour** (under 1% of bone tumours). Age 10-30.
- **Metaphysis of the proximal tibia** is the classic site (about one third around the knee), also distal femur, pelvis, feet.
- **Radiograph: eccentric, elongated, lobulated lytic metaphyseal lesion** with a **sclerotic scalloped ("bite-like") margin**, cortical expansion, **calcification uncommon** (unlike other cartilage lesions).
- **Histology:** lobules of **stellate and spindle cells in myxoid and chondroid stroma**, hypocellular centres with hypercellular periphery; **GRM1** rearrangement.
- **Differentials:** NOF (also eccentric metaphyseal, but cortical, multiloculated, children), ABC, fibrous dysplasia, chondrosarcoma (histological mimic).
- **Treatment: extended curettage with adjuvant and bone grafting**; recurrence about 15-25%; en bloc resection for expendable bones.

6. Chondrosarcoma

Epidemiology and classification

- **Second commonest primary malignant bone tumour** (about 20-30%). **Adults 40-70 years**, slight male predominance. Rare in children (consider mesenchymal subtype).

- **Sites: pelvis (ilium) commonest**, then **proximal femur**, **proximal humerus**, ribs and scapula. Flat bones and proximal limb bones (unlike enchondroma, which favours the hand).
- **Classification by origin:**
  - **Primary central (conventional)**: about 85-90%; arises de novo in the medulla.
  - **Secondary central**: from enchondroma (Ollier, Maffucci).
  - **Secondary peripheral**: from the cap of an osteochondroma (solitary or HME); usually low grade; treated by wide excision.
  - **Periosteal chondrosarcoma**: surface.
- **Subtypes:** dedifferentiated (about 10%), clear cell (about 2%), mesenchymal (about 2-3%).

Clinical and radiology

- **Pain (often deep, dull, nocturnal) is the key symptom**; mass in pelvic lesions often large at presentation; pathological fracture in high grade.
- **Radiograph and CT:** lytic expansile lesion with **rings-and-arcs calcification**, **deep endosteal scalloping**, **cortical thickening**, cortical destruction, periosteal reaction, soft tissue mass. A **purely lytic area within a calcified cartilage lesion suggests dedifferentiation** (target the biopsy there).
- **MRI:** lobulated, **very high T2 signal** (hyaline cartilage), peripheral and septal ("ring and arc") enhancement; defines extent and soft tissue mass.
- **Staging:** CT chest, bone scan or PET-CT. Biopsy from the most aggressive area (CT/MRI guided), at the treating centre.

Grading and prognosis

Grade	Histology	Metastasis	Approximate 10-year survival
ACT / grade 1	Low cellularity, mild atypia, abundant hyaline matrix	Rare	About 90%
Grade 2	More cellular, myxoid change, nuclear atypia, mitoses rare	About 10-30%	About 60-70%
Grade 3	Highly cellular, pleomorphic, mitoses, myxoid	Up to 70%	About 30-50%
Dedifferentiated	Low-grade cartilage **abruptly juxtaposed with high-grade non-cartilaginous sarcoma** (osteosarcoma, UPS, fibrosarcoma)	High	**5-year survival about 10-25%**

- **Grade is the most important prognostic factor**; others: margin status (local recurrence), axial site, size.
- Lungs are the commonest metastatic site.

Management

- **Conventional chondrosarcoma is resistant to chemotherapy and radiotherapy** (low mitotic rate, poor vascularity, abundant matrix), so **surgery is the mainstay**.
- **ACT of long bones (intracompartmental, no soft tissue extension): extended intralesional curettage with local adjuvant** (high-speed burr, phenol, liquid nitrogen cryotherapy, argon, or ethanol) and cement or graft; local recurrence low (about 5%), with better function than resection. Small asymptomatic lesions are increasingly managed by **active surveillance** (debated).

- **Grade 1 axial (pelvis, scapula, skull base), grade 2, grade 3, and secondary peripheral chondrosarcoma: wide en bloc resection**. Intralesional surgery leads to recurrence, and recurrent tumours can be of higher grade.
  - **Pelvic chondrosarcoma (Enneking-Dunham pelvic resection zones): I = ilium, II = periacetabular, III = pubis and ischium (obturator ring), IV = sacral extension**. Periacetabular (zone II) resections require reconstruction (custom or modular pelvic prosthesis, ice-cream-cone implants, allograft, hip transposition) or are left as a flail hip; **hindquarter amputation** if wide margins cannot otherwise be achieved.
- **Radiotherapy:** for incomplete resection, inoperable sites (skull base, spine) using high-dose **proton or carbon-ion therapy**; palliation.
- **Chemotherapy:** only for **mesenchymal** (Ewing-type regimens) and **dedifferentiated** chondrosarcoma (osteosarcoma-type regimens; benefit debated), and metastatic disease in trials. **IDH1 inhibitors (ivosidenib)** are under study for IDH-mutant tumours.
- **Follow-up:** long term (10 years or more) because of late local recurrence and lung metastases.

Other subtypes

Subtype	Key features	Treatment
Clear cell chondrosarcoma	Low grade; **epiphysis of proximal femur (about two thirds) or proximal humerus**; adults 25-50, males; well-defined lytic epiphyseal lesion; **mimics chondroblastoma** but in a skeletally mature patient; clear cells with osteoclast-like giant cells	**Wide en bloc resection** (curettage leads to high recurrence); late metastases possible, long follow-up

Subtype	Key features	Treatment
Mesenchymal chondrosarcoma	Young (10-30); jaw, ribs, spine, pelvis, and **extraskeletal** (meninges); **biphasic: undifferentiated small round blue cells with islands of hyaline cartilage**, haemangiopericytoma-like vessels; **HEY1-NCOA2** fusion; CD99 positive (differential: Ewing)	Wide resection + **chemotherapy** (Ewing-type) ± radiotherapy; poor prognosis, late metastases
Dedifferentiated chondrosarcoma	Older (50-70); pelvis, femur, humerus; **bimorphic** pattern; lytic aggressive component next to calcified cartilage; pathological fracture common	Wide resection or amputation ± chemotherapy; very poor prognosis
Periosteal chondrosarcoma	Surface of femur/humerus; usually **over 5 cm**; invades cortex	Wide excision; good prognosis (low grade)

Exam pearl: Chondrosarcoma: adult, pelvis commonest, painful lesion with rings-and-arcs calcification and deep endosteal scalloping. Chemo- and radio-resistant; grade is the key prognostic factor. Appendicular ACT = extended curettage + adjuvant; axial grade 1 and all grade 2/3 = wide resection. Clear cell CS = adult epiphysis of proximal femur (chondroblastoma look-alike). Mesenchymal CS = small round cells + cartilage, HEY1-NCOA2, chemotherapy-sensitive. Dedifferentiated = bimorphic, worst prognosis.

7. Synovial chondromatosis

- Classified with cartilage tumours (WHO 2020) as a benign, locally aggressive neoplasm of the synovium (**FN1-ACVR2A / FN1 rearrangements**).

- Monoarticular, adults 30-50, **knee commonest**, then hip, elbow, shoulder.
- Multiple cartilaginous nodules in the synovium that detach as loose bodies and may calcify (**multiple uniform-sized "rice-grain" ossified loose bodies**); erosion of bone on both sides of the joint.
- **Milgram stages: I** active intrasynovial disease, no loose bodies; **II** transitional, synovial nodules and loose bodies; **III** multiple loose bodies, quiescent synovium.
- **Treatment: loose body removal + synovectomy** (open or arthroscopic); recurrence about 10-20%. Malignant transformation to chondrosarcoma is rare (under about 5%), suspected with rapid recurrence or marrow invasion.
- Differentials: PVNS (haemosiderin, blooming on gradient-echo MRI, no calcification), secondary osteochondromatosis (loose bodies of varying size from OA).

8. Quick comparison table of cartilage tumours

Lesion	Age	Site	Discriminating clue
Osteochondroma	10-30	Metaphysis around knee	Corticomedullary continuity, points away from joint
Enchondroma	20-50	Hand phalanges, long bone medulla	Pathological fracture of proximal phalanx; rings-and-arcs
Chondroblastoma	10-25	**Epiphysis**, proximal humerus	Chicken-wire calcification, coffee-bean nuclei, oedema on MRI
Chondromyxoid fibroma	10-30	Proximal tibial metaphysis, eccentric	Scalloped sclerotic rim, little calcification

Lesion	Age	Site	Discriminating clue
Chondrosarcoma	40-70	Pelvis, proximal femur/humerus	Pain, deep endosteal scalloping, cortical thickening
Clear cell CS	25-50	Epiphysis proximal femur	Adult "chondroblastoma"
Mesenchymal CS	10-30	Jaw, ribs, extraskeletal	Small round cells + cartilage

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.